

Maci Kesler

mkesler1@kent.edu

Center for Nuclear Research

Department of Physics

Kent State University

Kent, OH USA 44240

Website: macink.github.io

Publication List

First Author Papers

- 2026  **M. Kesler and Z. Xu.** Mapping Gluon Distributions Through Deep Learning AI Models. *In preparation* (2026)
-  **M. Kesler, A. I. Sheikh, R. Ma, Z. Tu, T. Ullrich, and Z. Xu.** Projective Imaging of High-Energy Nuclei via Coherent Exclusive Vector Meson Production in Electron-Nucleus Collisions. *Phys. Lett. B* 875 (2026) 140585, *arXiv:2502.15596 [nucl-th]* (2025), DOI: 10.1016/j.physletb.2026.140585
- 2025  **M. Kesler, A. I. Sheikh, R. Ma, Z. Tu, T. Ullrich, and Z. Xu.** ePIC Analysis Note: Precision Extraction of Momentum Transfer in Diffractive Coherent Exclusive Vector Meson Production. *Zenodo* (2025), DOI: 10.5281/zenodo.17944278

Co-Author Papers

- 2026  **The ePIC Exclusive, Diffractive and Tagging Physics Working Group on behalf of the ePIC Collaboration.** Opportunities for Exclusive, Diffractive and Tagging Physics at the Electron-Ion Collider in the Early Science Period. *In preparation* (2026)
-  **ePIC Collaboration.** ePIC Early Science Report. *arXiv:2608.05377 [nucl-ex]* (2026), DOI: 10.48550/arXiv.2608.05377
-  **ePIC Collaboration.** The ePIC Detector Preliminary Technical Design Report. *Zenodo* (2026), DOI: 10.5281/zenodo.18271602

Collaboration Papers

-  **STAR Collaboration.** Energy Correlators within Jets in Transversely Polarized Proton-Proton Collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Rev. Lett.* 137 (2026) 11, 111902, *arXiv:2604.15546 [hep-ex]* (2026), DOI: 10.1103/fgsf-gx2q
-  **STAR Collaboration.** Inclusive Jet Cross Section in pp Collisions at $\sqrt{s_{NN}} = 200$ and 510 GeV. *arXiv:2603.28695 [hep-ex]* (2026), DOI: 10.48550/arXiv.2603.28695
-  **STAR Collaboration.** Probing Late-Stage Hadronic Interactions at High Baryon Density via K^{*0} Production in the RHIC Beam Energy Scan Program. *Phys. Lett. B* 876 (2026) 140389, *arXiv:2601.14884 [nucl-ex]* (2026), DOI: 10.1016/j.physletb.2026.140389
-  **STAR Collaboration.** Measurement of Fifth- and Sixth-Order Fluctuations of (Net-)Proton Number in Au + Au Collisions from Phase II of the Beam Energy Scan Program at RHIC. *Phys. Rev. C* 113 (2026) 5, L051901, *arXiv:2512.19352 [nucl-ex]* (2025), DOI: 10.1103/lq8c-9fr6
-  **STAR Collaboration.** Identified Charged Hadron Production in Au + Au Collisions at $\sqrt{s_{NN}} = 54.4$ GeV with the STAR Detector. *Phys. Rev. C* 113 (2026) 5, 054907, *arXiv:2512.06415 [nucl-ex]* (2025), DOI: 10.1103/4161-dflc
-  **STAR Collaboration.** Evidence of Spin-Interference Effects in Exclusive $J/\psi \rightarrow e^+e^-$ Photoproduction in Ultraperipheral Heavy-Ion Collisions. *Phys. Rev. Lett.* 136 (2026) 24, 242302, *arXiv:2512.02865 [nucl-ex]* (2025), DOI: 10.1103/tcdb-ldh8
-  **STAR Collaboration.** First Observation of Deuteron- Λ Correlations at RHIC. *Phys. Rev. Lett.* 136 (2026) 24, 242303, *arXiv:2511.15493 [nucl-ex]* (2025), DOI: 10.1103/3m26-5y83

Publication List (continued)

- **STAR Collaboration.** Study of Central Exclusive Production of $\pi^+\pi^-$, K^+K^- , and $p\bar{p}$ Pairs in Proton-Proton Collisions at $\sqrt{s_{NN}} = 510$ GeV with the STAR Detector at RHIC. *JHEP* 07 (2026) 143, *arXiv:2510.27482 [hep-ex]* (2025), DOI: 10.1007/JHEP07(2026)143
- **STAR Collaboration.** Measurement of Transverse Polarization of Λ and $\bar{\Lambda}$ Hyperons Inside Jets in Unpolarized Proton-Proton Collisions at $\sqrt{s_{NN}} = 200$ GeV. *JHEP* 06 (2026) 050, *arXiv:2509.17487 [hep-ex]* (2025), DOI: 10.1007/JHEP06(2026)050
- **STAR Collaboration.** Observation of Charmonium Sequential Suppression in Heavy-Ion Collisions at the Relativistic Heavy Ion Collider. *Phys. Rev. Lett.* 136 (2026) 12, 122302, *arXiv:2509.12842 [nucl-ex]* (2025), DOI: 10.1103/ghzh-kv8z
- **STAR Collaboration.** Measurement of Inclusive J/ψ Production in Au + Au Collisions at $\sqrt{s_{NN}} = 54.4$ GeV at STAR. *Phys. Lett. B* 877 (2026) 140405, *arXiv:2506.20962 [nucl-ex]* (2025), DOI: 10.1016/j.physletb.2026.140405
- **STAR Collaboration.** Measuring Spin Correlation Between Quarks During QCD Confinement. *Nat.* 650 (2026) 8100, 66–71, *arXiv:2506.05499 [hep-ex]* (2025), DOI: 10.1038/s41586-025-09920-0
- **STAR Collaboration.** Measurement of the Longitudinal Flow-Plane Decorrelation Using Multiplane Cumulants in $\sqrt{s_{NN}} = 200$ GeV Au + Au, Ru + Ru, and Zr + Zr Collisions. *Phys. Rev. C* 113 (2026) 2, 024908, *arXiv:2506.02330 [nucl-ex]* (2025), DOI: 10.1103/75tf-ljtk
- **STAR Collaboration.** Search for the Chiral Magnetic Effect Through Beam Energy Dependence of Charge Separation Using Event Shape Selection. *Phys. Rev. C* 113 (2026) 1, 014912, *arXiv:2506.00278 [nucl-ex]* (2025), DOI: 10.1103/pj7s-9ym7
- **STAR Collaboration.** Measurement of Medium-Induced Acoplanarity in Central Au – Au and pp Collisions at $\sqrt{s_{NN}} = 200$ GeV Using Direct-Photon+Jet and π^0 +Jet Correlations. *Phys. Rev. C* 113 (2026) 1, 014902, *arXiv:2505.05789 [nucl-ex]* (2025), DOI: 10.1103/k29c-d5ry
- 2025 ■ **STAR Collaboration.** Energy Independence of the Collins Asymmetry in p^+p Collisions. *Phys. Rev. Lett.* 135 (2025) 26, 261902, *arXiv:2507.16355 [hep-ex]* (2025), DOI: 10.1103/jgsw-zhny
- **STAR Collaboration.** Hyperon Global Polarization in Isobar Ru + Ru and Zr + Zr Collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Lett. B* 870 (2025), 139891, *arXiv:2505.05046 [nucl-ex]* (2025), DOI: 10.1016/j.physletb.2025.13989
- **STAR Collaboration.** Onset of Constituent Quark Number Scaling in Heavy-Ion Collisions at RHIC. *Phys. Rev. Lett.* 135 (2025) 7, 072301, *arXiv:2504.02531 [nucl-ex]* (2025), DOI: 10.1103/2qhx-cp79
- **STAR Collaboration.** Precision Measurement of Net-Proton-Number Fluctuations in Au+Au Collisions at RHIC. *Phys. Rev. Lett.* 135 (2025) 14, 142301, *arXiv:2504.00817 [nucl-ex]* (2025), DOI: 10.1103/9l69-2d7p
- **STAR Collaboration.** Measurement of Two-Point Energy Correlators within Jets in $p + p$ Collisions at $\sqrt{s} = 200$ GeV. *Phys. Rev. Lett.* 135 (2025) 11, 111901, *arXiv:2502.15925 [hep-ex]* (2025), DOI: 10.1103/wv2t-dkgn
- **STAR Collaboration.** Measurements of v States Production in $p + p$ Collisions at $\sqrt{s} = 500$ GeV with STAR: Cross Sections, Ratios, and Multiplicity Dependence. *Phys. Rev. D* 112 (2025) 3, 032004, *arXiv:2502.03769 [hep-ex]* (2025), DOI: 10.1103/bsyx-qtjp
- **STAR Collaboration.** Temperature Measurement of Quark-Gluon Plasma at Different Stages. *Nat. Commun.* 16 (2025) 1, 9098, *arXiv:2402.01998 [nucl-ex]* (2024), DOI: 10.1038/s41467-025-63216-5